HopTF: Sequence-Conditioned Associative Retrieval for Hypothesis-Generating Transcription Factor Perturbation Modeling

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Abstract

Transcription factor (TF) overexpression can induce coordinated changes across gene regulatory programs, but existing single-cell perturbation models often represent perturbations as categorical gene identifiers, learned embeddings, or graph-derived features. These representations are limited when the perturbation is a TF isoform with sequence-specific regulatory potential and when the goal is to generate hypotheses about which regulatory loci mediate the response. We introduce HopTF, a biophysically motivated, sequence-conditioned perturbation model that frames TF response prediction as associative retrieval over a genome-indexed memory bank. HopTF represents each TF perturbation using a protein language model embedding of the TF isoform sequence. This TF embedding acts as a query over AlphaGenome-derived genomic locus embeddings, optionally modulated by cell-state or cluster-specific accessibility context. The resulting Hopfield/attention retrieval distribution defines a soft activation pattern over candidate regulatory loci, and the pooled representation μ_p conditions a flow model that predicts the transport from control-cell latent states to TF-perturbed latent states. The model is trained end-to-end from aggregate perturbation-response supervision, without locus-level labels, giving the problem a weakly supervised multiple-instance structure. We position HopTF as a hypothesis-generating model rather than a definitive binding-site predictor: high-retrieval loci are interpreted as model-prioritized regulatory-context hypotheses that require validation through motif enrichment, accessibility consistency, target-gene proximity, and controls for protein length, ORF length, recovered cell count, and shuffled sequence representations.

In a PCA-space benchmark on the TF Atlas local subsample, we asked whether a TF sequence representation still helps to predict the state of the perturbed cell after the model already sees protein length, ORF length, and recovered cell count. Using only those three quantities gave MSE 6.5996. Adding the ESM-C-derived representation after projection into AlphaGenome key space reduced MSE to 6.2879, a 4.7% reduction. Raw protein embeddings alone did not outperform the model using only protein length, ORF length, and recovered cell count, and the DBP-specialized ESM-DBP encoder underperformed ESM-C in this benchmark. These results support HopTF as a useful retrieval-and-diagnostic prototype, but they also show that measured non-sequence variables strongly predict the state of the perturbed cell and should not be ignored.

1 Introduction

Modeling how transcription factors reshape cellular state is a central problem in systems biology. TFs regulate large gene programs by binding regulatory DNA, interacting with cofactors, and altering transcriptional output. This makes them natural targets for directed differentiation, cell-state engineering, and regulatory network inference. The TF Atlas of directed differentiation profiled overexpression responses for all annotated human TF isoforms in H1 human embryonic stem cells at single-cell resolution, providing a large-scale resource for learning TF-induced transcriptional responses (Joung et al., 2023). However, the dataset is fundamentally aggregate: we observe the downstream expression phenotype for a TF perturbation, but not the individual regulatory loci responsible for mediating that response.

This observation motivates the core modeling question of this project: can a TF isoform sequence be used as a molecular cue to retrieve regulatory contexts from the genome, and can the retrieved context condition prediction of the TF-induced cell-state shift? HopTF addresses this question by combining three pretrained biological representation layers with a differentiable associative retrieval bottleneck. First, genomic loci are represented by AlphaGenome-derived regulatory embeddings, giving a genome-indexed memory bank of candidate regulatory contexts. Second, TF isoforms are represented by protein language model embeddings derived from their amino-acid sequences. Third, single-cell expression profiles are mapped into a latent cell-state space using a multimodal single-cell encoder. A Hopfield-style retrieval module then uses the TF embedding, optionally modulated by cell state, to produce an activation distribution over genomic loci. The weighted pooled locus representation μ_p conditions a continuous flow model that transports control-cell states toward TF-perturbed states.

The conceptual contribution is not the use of a softmax or attention operation alone. Modern Hopfield networks and transformer attention are mathematically related (Ramsauer et al., 2021; Vaswani et al., 2017), and generic attention layers are now common across biological deep learning. The distinction is that HopTF assigns the retrieval operation a specific biological role. The query is a TF isoform sequence embedding; the memory slots are genomic regulatory locus embeddings; the retrieved state is a regulatory-context representation; and the downstream target is the TF-conditioned perturbation transport field. This decomposition gives the retrieval distribution biological semantics and makes it a primary object of analysis rather than an incidental hidden layer.

We therefore frame HopTF as a sequence-conditioned associative retrieval model for hypothesis-generating TF perturbation prediction. The model is trained using only aggregate perturbation-response supervision. It is not told which loci mediate the TF response. It must instead learn a retrieval distribution over candidate loci whose pooled representation improves prediction of the observed transcriptional response. This induces a weakly supervised multiple-instance structure: for a given cell state, the genome, or an accessibility-filtered subset of the genome in masked variants, forms a bag of candidate regulatory loci, while the TF isoform embedding acts as a query over this bag.

Claim boundary. We treat the associative-memory framing as a computational abstraction. We do not claim that the genome is literally a Hopfield memory, nor that high-retrieval loci are validated TF binding sites. Instead, high-retrieval loci are model-prioritized regulatory-context hypotheses. The appropriate biological validation is diagnostic: motif enrichment, accessibility consistency, overlap with external ChIP-seq or perturbation evidence when available, proximity to affected target genes, and robustness to controls based on protein length, ORF length, recovered cell count, and shuffled sequence representations.

Summary of intended contributions.

1. We formulate TF perturbation prediction as sequence-conditioned associative retrieval over a genome-indexed regulatory memory bank.
2. We connect the retrieval module to weakly supervised multiple-instance learning, where locus-level mediators are latent and supervision is available only at the aggregate perturbation-response level.
3. We combine the retrieved regulatory-context representation with a conditional flow field for predicting cell-state transport under TF overexpression.
4. We propose retrieval diagnostics and controls that make the model useful for hypothesis generation rather than opaque prediction of perturbed-cell state.

Figure placeholder: Overview of HopTF

Schematic to create. Left: TF isoform amino-acid sequence is encoded by a protein language model and projected into query space. Middle: genomic loci are encoded by AlphaGenome into a memory bank of keys and values. Cell state optionally modulates retrieval through accessibility or cluster-specific bias. Right: retrieved representation μ_p conditions a flow model that transports control-cell latent states to TF-perturbed latent states. Add a separate panel showing retrieval weights over loci as interpretable activation patterns.

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2 Background and Related Work

2.1 Associative memory, Hopfield retrieval, and attention

Classical Hopfield networks introduced content-addressable associative memory: stored patterns can be retrieved from partial or noisy cues through energy-based dynamics (Hopfield, 1982). Dense associative memory generalized this idea to higher-capacity energy functions capable of storing and retrieving many more patterns (Krotov and Hopfield, 2016). Modern Hopfield networks extend associative memory to continuous states and show that the update rule is equivalent to the attention mechanism used in transformers (Ramsauer et al., 2021; Vaswani et al., 2017). This connection supports a retrieval-centric interpretation of attention: a query retrieves stored patterns through similarity-weighted pooling.

This perspective has already been useful in biological multiple-instance learning. DeepRC uses transformer-like attention, equivalently modern Hopfield retrieval, to classify immune repertoires from extremely large bags of immune receptor sequences with sparse disease-associated witnesses (Widrich et al., 2020). The analogy to HopTF is direct but not identical. In immune repertoire classification, the model must identify a small set of disease-relevant sequences from a massive repertoire. In HopTF, the model must identify TF-compatible regulatory contexts from a large set of genomic loci using only aggregate transcriptional response supervision.

2.2 Single-cell perturbation response models

A growing literature models transcriptional responses to genetic or chemical perturbations in single cells. GEARS represents perturbations using gene embeddings and gene-gene relationship graphs to predict outcomes for held-out genetic perturbations (Roohani et al., 2023). CellOT learns maps between control and perturbed cell-state distributions using optimal transport (Bunne et al., 2023). scGPT and related single-cell foundation models use transformer-style pretraining over gene expression profiles for cell representation learning and downstream perturbation tasks (Cui et al., 2024). Recent benchmarking work has emphasized that simple comparisons can be difficult to beat in single-cell perturbation prediction, making controls for dataset structure and rigorous split design essential (Ahlmann-Eltze et al., 2025).

These methods are important comparators, but most do not explicitly represent a TF perturbation by its amino-acid sequence, retrieve regulatory loci, and expose a locus-level activation distribution. HopTF is therefore positioned less as a generic perturbed-state predictor and more as a model for sequence-conditioned hypothesis generation about TF-compatible regulatory contexts.

2.3 Mechanistic and regulatory-network approaches

CellOracle performs in silico TF perturbation by constructing gene regulatory networks from single-cell multi-omics data and simulating downstream expression changes after TF perturbation (Kamimoto et al., 2023). SCENIC+ infers enhancer-driven gene regulatory networks by identifying candidate enhancers, enriched TF motifs, and links between TFs, enhancers, and target genes (Bravo González-Blas et al., 2023). These frameworks provide mechanistic biological interpretability through motifs, regulatory regions, and network structure. However, they generally rely on motif libraries, enhancer-gene linking, and regression or network propagation rather than protein-sequence-conditioned differentiable retrieval.

HopTF is complementary to these approaches. It does not replace motif- or ChIP-based validation. Instead, it produces a learned retrieval distribution over loci that can be compared against motif enrichment, accessibility, target gene annotations, and external binding evidence.

2.4 Protein-aware TF-DNA binding models

TransBind and TFBindFormer are close protein-aware TF-DNA binding baselines because they condition DNA sequence predictions on TF protein representations through cross-attention (Basnet et al., 2026; Liu et al., 2026). These models are strong comparators for protein-conditioned binding prediction on local genomic sequence windows. The distinction is the modeling level. TransBind and TFBindFormer predict TF-DNA binding over local sequence bins or ChIP-style labels; HopTF retrieves from a genome-indexed regulatory memory and uses the retrieved state to condition a downstream cell-state perturbation model. In other words, HopTF is not a local binding classifier. It is a weakly supervised model of TF sequence, regulatory context, and aggregate perturbation response.

2.5 Sequence-conditioned transport

STRAND is a close conceptual neighbor because it models single-cell perturbation responses using sequence-conditioned transport (Fu et al., 2026). STRAND represents a perturbation by encoding regulatory DNA sequence at the genomic locus of intervention and uses that sequence representation to parameterize transport from control to perturbed cell states. HopTF differs in the direction of conditioning: the perturbation cue is the TF protein isoform sequence, and the model retrieves from a genome-wide regulatory memory bank before conditioning the flow field.

3 Problem Formulation

Let $x \in \mathbb{R}^G$ denote a single-cell expression vector over G genes. Let p index a TF perturbation, and let s_p denote the amino-acid sequence of the perturbed TF isoform. The training data consist of control cells and TF-perturbed cells. For each TF p, we observe samples from a control distribution P_0 and a perturbed distribution P_p, but we do not observe which genomic loci mediate the perturbation response.

We encode cells into a latent space using a cell encoder E_{cell}, such as SCARF:

$$z = E_{\text{cell}}(x) \in \mathbb{R}^{d_z}. \tag{1}$$

The downstream goal is to learn a conditional vector field that transports control-cell latents z_0 toward TF-perturbed latents $z_1 \sim P_p$:

$$\frac{dz_t}{dt} = v_\theta(z_t, t, z_0, \mu_{0p}), \qquad t \in [0, 1], \tag{2}$$

where μ_{0p} is a TF- and cell-state-conditioned perturbation representation produced by associative retrieval over genomic loci.

The key latent variable in the model is the locus activation distribution

$$\alpha_{0pi} = P_\theta(i \mid z_0, s_p), \qquad i = 1, \ldots, N, \tag{3}$$

where N is the number of candidate genomic loci. The model is not trained with labels for α_{0pi}. Instead, α_{0pi} is induced by the perturbation transport objective. This is the weakly supervised multiple-instance structure: the bag is the candidate locus set, the query is the TF isoform sequence, and the bag-level target is the aggregate transcriptional response.

4 Method

4.1 Genome-indexed regulatory memory

We partition the genome into N candidate loci $\{g_i\}_{i=1}^N$. A locus may be defined as a fixed-width genomic window, an accessible peak, a promoter/enhancer candidate, or a gene-linked regulatory region. In the base model, each locus is encoded using a frozen genome foundation model:

$$h_i^{\text{AG}} = E_{\text{AG}}(g_i) \in \mathbb{R}^{d_{AG}}, \tag{4}$$

where E_{AG} denotes AlphaGenome. AlphaGenome provides a regulatory sequence prior because it predicts functional genomic tracks across modalities including expression, accessibility, histone marks, TF binding, chromatin contacts, and splicing-related outputs (Avsec et al., 2026).

HopTF constructs a task-adapted key-value memory from these pretrained locus embeddings:

$$k_i = W_K h_i^{\mathrm{AG}}, \tag{5}$$

$$v_i = W_V h_i^{\mathrm{AG}}. \tag{6}$$

Here k_i is used for TF-locus compatibility scoring, while v_i is the representation retrieved and pooled by the model. A tied-memory variant sets $W_K = W_V$ or directly uses $k_i = v_i$. An untied variant allows the scoring representation and retrieved representation to differ.

Placeholder. Implementation choice to finalize: define the exact locus unit. Options: fixed-width windows, ATAC peaks, promoter/enhancer candidates, gene-linked windows, or AlphaGenome context blocks. State final N, window length L, genome build, overlap/stride, and whether coding regions are excluded or treated separately.

4.2 TF protein sequence query

For each TF isoform p, we compute a protein language model embedding from its amino-acid sequence:

$$e_p = E_{\mathrm{prot}}(s_p), \tag{7}$$

where E_{prot} is ESM-C or another protein language model. A learned projection maps this embedding into the associative retrieval space:

$$q_p = W_Q e_p \in \mathbb{R}^d. \tag{8}$$

This query represents the sequence-derived molecular identity of the perturbed TF isoform. Unlike perturbation-ID embeddings, this representation can in principle distinguish isoforms, sequence variants, domain truncations, or mutants if their protein sequences are provided.

In the current implementation, ESM-C uses the 600M-parameter model with mean pooling over non-special tokens, producing a frozen 1152-dimensional isoform embedding. Experiment 1 also tested ESM-DBP, a DBP-specialized ESM2-style 650M model from ESM-DBP, with mean non-special-token pooling into a frozen 1280-dimensional embedding. The ESM-DBP run embedded 3264 of 3548 vocabulary entries; 284 entries had no local amino-acid sequence and 139 sequences were truncated to the 1022-token model limit. Missing-sequence rows are retained in the aligned vocabulary with a mask, so downstream baselines can distinguish unavailable sequence information from real zero-valued embeddings.

4.3 Cell-state modulation of retrieval

The retrieval distribution may be global, cluster-specific, or cell-state-specific. The most conservative base model uses no cell-state-specific mask:

$$r_{pi} = \beta q_p^\top k_i. \tag{9}$$

This retrieves TF-compatible loci from a genome-wide memory bank.

A cell-state-conditioned version adds a retrieval bias or mask derived from the cell latent z_0:

$$r_{0pi} = \beta q_p^\top k_i + b_\eta(z_0, h_i^{\mathrm{AG}}), \tag{10}$$

where b_η may represent a learned cell-state/locus compatibility score, a cluster-specific accessibility prior, or a log-accessibility mask. If $m_i(z_0) \in \{0,1\}$ denotes a hard accessibility mask, this can be written as

$$r_{0pi} = \begin{cases} \beta q_p^\top k_i, & m_i(z_0) = 1, \\ -\infty, & m_i(z_0) = 0. \end{cases} \tag{11}$$

In practice, hard cell-specific masks may be unstable or unrealistic if accessibility is not observed for every target cell. A cluster-specific or cell-type-specific accessibility bias may be a more robust intermediate design:

$$r_{cpi} = \beta q_p^\top k_i + \lambda A_{ci}, \tag{12}$$

where c is a cluster in SCARF latent space and A_{ci} is the average accessibility or availability prior for locus i in cluster c.

> **Placeholder.** Implementation choice to finalize: specify whether the base experiment uses no mask, cluster-specific accessibility bias, or cell-specific accessibility estimates. If accessibility is predicted from SCARF, state how predicted accessibility is calibrated and mapped onto AlphaGenome loci.

4.4 Associative retrieval over genomic loci

Given retrieval scores r_{0pi}, HopTF computes a soft activation pattern over loci:

$$\alpha_{0pi} = \frac{\exp(r_{0pi})}{\sum_{j=1}^{N} \exp(r_{0pj})}. \tag{13}$$

The retrieved regulatory-context representation is

$$\mu_{0p} = \sum_{i=1}^{N} \alpha_{0pi} v_i. \tag{14}$$

In matrix form, with $K \in \mathbb{R}^{N\times d}$ and $V \in \mathbb{R}^{N\times d_v}$, the unmasked version is

$$\alpha_p = \mathrm{softmax}(\beta K q_p), \qquad \mu_p = V^\top \alpha_p. \tag{15}$$

If values are tied to keys, $V = K$, recovering the simpler expression

$$\mu_p = K^\top \mathrm{softmax}(\beta K q_p). \tag{16}$$

This is the associative-memory component of HopTF. Genomic loci are stored patterns, the TF embedding is the retrieval cue, and μ_{0p} is the retrieved regulatory-context prototype used for perturbation prediction.

The temperature parameter β controls retrieval sharpness. Low β yields diffuse averaging across many loci, while high β concentrates mass on a smaller set of loci. We treat β as an inverse-temperature parameter. It may be biologically suggestive of binding selectivity or dosage-dependent sharpness, but we do not assume a direct quantitative equivalence to TF concentration without dose-calibrated data.

4.5 Conditional flow field for perturbation response

The retrieved representation μ_{0p} conditions a time-dependent vector field in cell-state latent space:

$$\frac{dz_t}{dt} = v_\theta(z_t, t, z_0, \mu_{0p}). \tag{17}$$

During training, we use conditional flow matching with minibatch optimal transport couplings (Tong et al., 2024a,b). For a TF perturbation p, let (z_0, z_1) be a paired sample from a minibatch

OT coupling between control latents and perturbed latents. For $t \sim \text{Uniform}(0,1)$, define a linear interpolation

$$z_t = (1-t)z_0 + tz_1, \tag{18}$$

with target velocity

$$u_t = z_1 - z_0. \tag{19}$$

The flow matching objective is

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{p,(z_0,z_1),t}\left[\|v_\theta(z_t, t, z_0, \mu_{0p}) - (z_1 - z_0)\|_2^2\right]. \tag{20}$$

At inference, given a control cell x_0, we compute $z_0 = E_{\text{cell}}(x_0)$, retrieve μ_{0p}, and integrate the learned ODE from $t = 0$ to $t = 1$ to obtain a predicted perturbed latent $\hat{z}_1$. This latent can be decoded back to expression space if a decoder is available.

Placeholder. Implementation choice to finalize: specify whether the model predicts final perturbed-cell latent states, latent velocity fields, decoded gene expression, or all three. State whether SCARF is frozen, whether a decoder is used, and how the latent predictions are mapped to gene-level evaluation metrics.

4.6 End-to-end differentiability and weak supervision

The retrieval module is differentiable from retrieval scores through the downstream flow objective. Therefore, even though the model does not observe locus-level labels, gradients from the perturbation-response loss can shape the TF query projection, locus key/value projections, cell-state retrieval bias, and downstream flow model:

$$\mathcal{L}_{\text{CFM}} \to v_\theta \to \mu_{0p} \to \alpha_{0pi} \to (q_p, k_i, v_i, b_\eta). \tag{21}$$

If AlphaGenome, ESM-C, or SCARF are frozen, gradients do not update those foundation model parameters. The base framing is therefore not full fine-tuning of AlphaGenome. Instead, HopTF learns a task-adapted associative regulatory memory initialized by AlphaGenome-derived locus embeddings. A natural extension is to allow adapter or residual updates to the key/value memory slots:

$$k_i = W_K h_i^{\text{AG}} + \Delta k_i, \tag{22}$$

$$v_i = W_V h_i^{\text{AG}} + \Delta v_i, \tag{23}$$

with regularization on Δk_i and Δv_i to preserve the pretrained regulatory prior.

Placeholder. Implementation choice to finalize: specify which components are trained, frozen, or adapter-tuned. If residual memory updates are used, report the regularization strength and whether retrieval diagnostics are robust to these updates.

4.7 Retrieval diagnostics

Because the retrieval distribution is an object of interpretation, not just a hidden layer, we evaluate it directly. For each cell/TF pair, define retrieval entropy

$$H(\alpha_{0p}) = -\sum_i \alpha_{0pi} \log \alpha_{0pi} \tag{24}$$

and effective number of retrieved loci

$$N_{\text{eff}}(\alpha_{0p}) = \exp(H(\alpha_{0p})). \tag{25}$$

We also compute top-k mass

$$M_k(\alpha_{0p}) = \sum_{i \in \text{TopK}(\alpha_{0p})} \alpha_{0pi}. \tag{26}$$

These metrics quantify whether retrieval is diffuse, subset-like, or collapsed. Biological diagnostics include motif enrichment among top-retrieved loci, accessibility consistency, overlap with external ChIP-seq peaks when available, and proximity or linkage to genes with altered expression after TF overexpression.

5 Experimental Design

5.1 Datasets

Primary perturbation data. The main training data are planned to come from the TF Atlas of directed differentiation, which profiles TF isoform overexpression in H1 hESCs at single-cell resolution (Joung et al., 2023). Each perturbation is associated with a TF isoform sequence and a set of observed perturbed cells. Control cells define the source distribution for flow matching.

> **Placeholder.** Fill in: number of control cells, number of perturbation cells, number of TF isoforms after filtering, number of genes retained, preprocessing steps, quality-control thresholds, train/validation/test splits, and whether perturbations are split by TF family, held-out isoform, held-out domain, or random TF.

Genomic locus data. Candidate loci are represented by AlphaGenome-derived embeddings. Loci may be defined by fixed genomic windows, accessible peaks, promoter/enhancer annotations, or gene-linked windows.

> **Placeholder.** Fill in: genome build, locus definition, window size, stride, number of loci N, AlphaGenome embedding extraction procedure, whether embeddings are pooled across tracks/layers, and whether locus values include annotations beyond AlphaGenome.

Cell-state representation. Cells are embedded into a latent space using SCARF or another single-cell encoder. In cell-contextualized retrieval variants, cell state may define an accessibility prior or cluster identity.

> **Placeholder.** Fill in: SCARF model version, input modality, latent dimension, whether RNA-only or RNA+ATAC is used, clustering method, number of clusters, and whether accessibility is measured or predicted.

5.2 Evaluation splits

We propose several levels of generalization:

1. **Random held-out cells within seen TFs:** tests whether the flow model fits observed perturbation distributions.

2. **Held-out TF isoforms:** tests sequence-conditioned generalization to TFs not seen in training.
3. **Held-out TF families or domains:** tests harder extrapolation across DNA-binding families.
4. **Held-out cell-state clusters:** if multiple cell states are used, tests whether retrieval and transport generalize across cellular context.

Placeholder. Choose final split strategy. For a class report, include at least one perturbation-level split rather than only a random cell-level split, because random cell splits can overstate performance.

5.3 Comparisons and controls

The following comparisons are intended to separate biological signal from architecture and dataset structure:

Table 1: Planned comparisons and their purpose.

Comparison	Description	Question tested
Control / no-perturbation	Predict no movement from control latent.	Is any perturbation modeling learned?
Average TF effect	Use the mean perturbation shift across training TFs.	Are gains TF-specific or just average response?
TF ID embedding	Replace protein sequence with a learned TF identity embedding.	Does protein sequence add value?
ESM-only flow	Condition flow on TF protein embedding without genomic memory retrieval.	Does locus retrieval add value beyond protein sequence?
Random / shuffled memory	Replace AlphaGenome loci with random or shuffled locus embeddings.	Does regulatory memory carry useful signal?
Shuffled TF sequence	Query memory using shuffled or mismatched TF embeddings.	Are retrieval weights TF-specific?
No cell-state bias	Remove cell-state or cluster-specific accessibility bias.	Does cell-contextualization matter?
Length / ORF / cell-count inputs	Predict from sequence length, ORF length, and perturbation cell count.	How much can these measured quantities explain?

Placeholder. Insert final implemented comparisons. Mark any planned but not completed comparisons clearly.

5.4 Perturbed-cell state prediction metrics

We evaluate perturbation prediction in latent and gene-expression space:

- latent MSE or cosine distance between predicted and observed perturbed latents;
- Pearson/Spearman correlation between predicted and observed mean expression shifts;
- correlation restricted to differentially expressed genes;

- distributional metrics such as MMD or Wasserstein distance between predicted and observed perturbed-cell distributions;
- top-k differential-expression recovery or direction accuracy.

Placeholder. Define final primary metric. Suggested: use one main metric for perturbed-cell state prediction and separate diagnostics for retrieval; avoid overloading the paper with too many metrics.

5.5 Retrieval and biological validation metrics

We evaluate the locus activation distribution independently of perturbed-cell state prediction:

- retrieval entropy $H(\alpha)$, effective number of loci N_{eff}, and top-k mass;
- beta sensitivity curves showing diffuse, subset, and concentrated retrieval regimes;
- motif enrichment among top-retrieved loci using matched GC/accessibility/length backgrounds;
- overlap with external ChIP-seq peaks for the same or related TFs when available;
- enrichment near genes whose expression changes after TF overexpression;
- stability of retrieved loci across nearby cells or clusters.

Placeholder. Specify motif enrichment tool, motif database, background construction, top-k thresholds, and multiple-testing correction.

6 Results

6.1 Predicting the state of the perturbed cell

We first evaluated prediction of the perturbed-cell state in a conservative PCA-latent benchmark using grouped held-out-gene splits. The purpose of this experiment was not to claim final transport performance, but to ask whether a TF sequence representation still helps to predict the state of the perturbed cell after the model already sees protein length, ORF length, and recovered cell count. The sequence-derived query was produced by projecting the frozen protein embedding into the AlphaGenome key space and retrieving against the same AlphaGenome-derived memory used by the rest of the model.

Table 2: Experiment 1 prediction of the perturbed-cell state on grouped held-out-gene splits. Lower MSE is better.

Feature set	MSE	Median row MSE	Mean cosine similarity
Protein length + ORF length + recovered cell count	6.5996	1.7827	0.3970
ESM-C projected into AlphaGenome key space only	6.8806	1.9395	0.3395
Raw ESM-C embedding	7.3078	2.7258	0.3024
ESM-DBP projected into AlphaGenome key space only	7.9615	3.0910	0.2711
Raw ESM-DBP embedding	8.4732	3.5707	0.2537
Shuffled ESM-C features	8.5951	3.1644	0.0491
Shuffled ESM-DBP features	9.0622	3.3394	0.0571
Training-set mean perturbed-cell state	37.6192	40.6827	-0.4740

The model using only protein length, ORF length, and recovered cell count was difficult to beat. Raw ESM-C and raw ESM-DBP embeddings performed worse than those three quantities alone. The ESM-C-derived representation after projection into AlphaGenome key space was stronger than raw ESM-C, but by itself it still did not beat the model using only protein length, ORF length, and recovered cell count. This motivates the stricter evaluation below, where sequence features are added after the model already sees those three quantities.

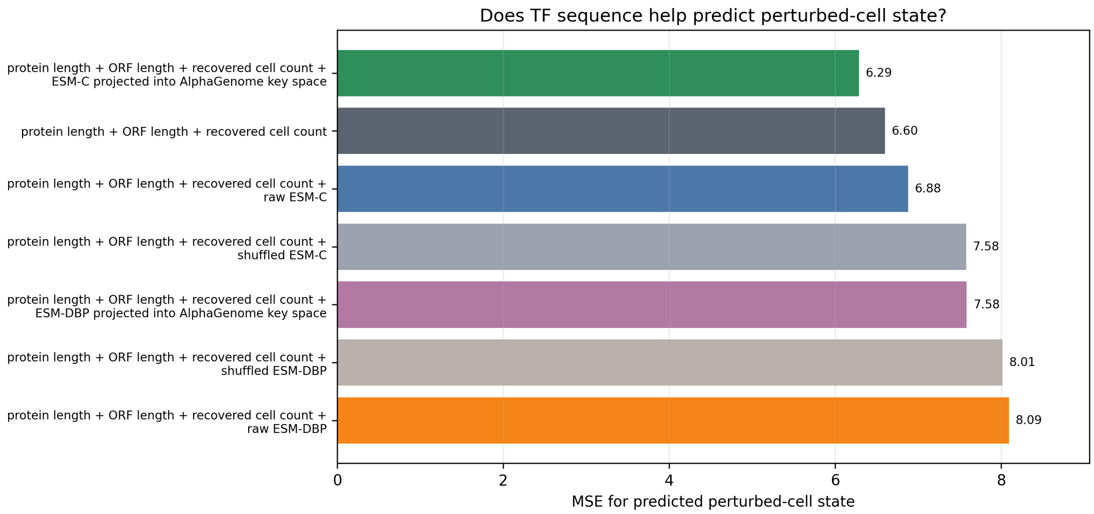

Figure 2: MSE for predicting the perturbed-cell state. The ESM-C-derived representation after projection into AlphaGenome key space modestly improves over using only protein length, ORF length, and recovered cell count. ESM-DBP does not improve this benchmark.

6.2 Ablation studies

Experiment 1B compared the generic ESM-C protein encoder with the DBP-specialized ESM-DBP encoder under the same splits and the same protein-length, ORF-length, and recovered-cell-count inputs. The ESM-DBP embedding generator used the downloaded ESM-DBP checkpoint from the shared cluster path and produced aligned isoform embeddings before fitting the projection into AlphaGenome key space. The ESM-DBP projection required a higher inverse-temperature setting than the default: with $\beta = 30$, 2500 steps, and 64 overfit examples, it reached 100% top-1 retrieval accuracy in the projection smoke test.

Table 3: Overall encoder ablation. Percent change is relative to using only protein length, ORF length, and recovered cell count. Negative values indicate lower MSE.

Feature set	MSE	MSE change vs three-input model	Interpretation
Protein length + ORF length + recovered cell count	6.5996	0.0%	Reference for this comparison
Protein length + ORF length + recovered cell count + ESM-C projected into AlphaGenome key space	6.2879	-4.7%	Lowest MSE in this run
Protein length + ORF length + recovered cell count + raw ESM-C	6.8774	+4.2%	Raw sequence embedding does not help after those three quantities
Raw ESM-C only	7.3078	+10.7%	Weaker than using only protein length, ORF length, and recovered cell count
Protein length + ORF length + recovered cell count + ESM-DBP projected into AlphaGenome key space	7.5832	+14.9%	DBP-specialized representation underperforms ESM-C
Protein length + ORF length + recovered cell count + raw ESM-DBP	8.0926	+22.6%	DBP-specialized raw embedding underperforms ESM-C
Protein length + ORF length + recovered cell count + shuffled ESM-C	7.5761	+14.8%	Negative control worsens as expected
Protein length + ORF length + recovered cell count + shuffled ESM-DBP	8.0145	+21.4%	Negative control worsens as expected

The useful signal in this run appears only after taking the ESM-C embedding and projecting it into the AlphaGenome key space. The raw protein embeddings do not improve prediction of the perturbed-cell state, and ESM-DBP does not replace ESM-C under this comparison. This argues against treating generic protein embeddings as sufficient and supports keeping the retrieval step as the main object of analysis, while also showing that the gains are modest.

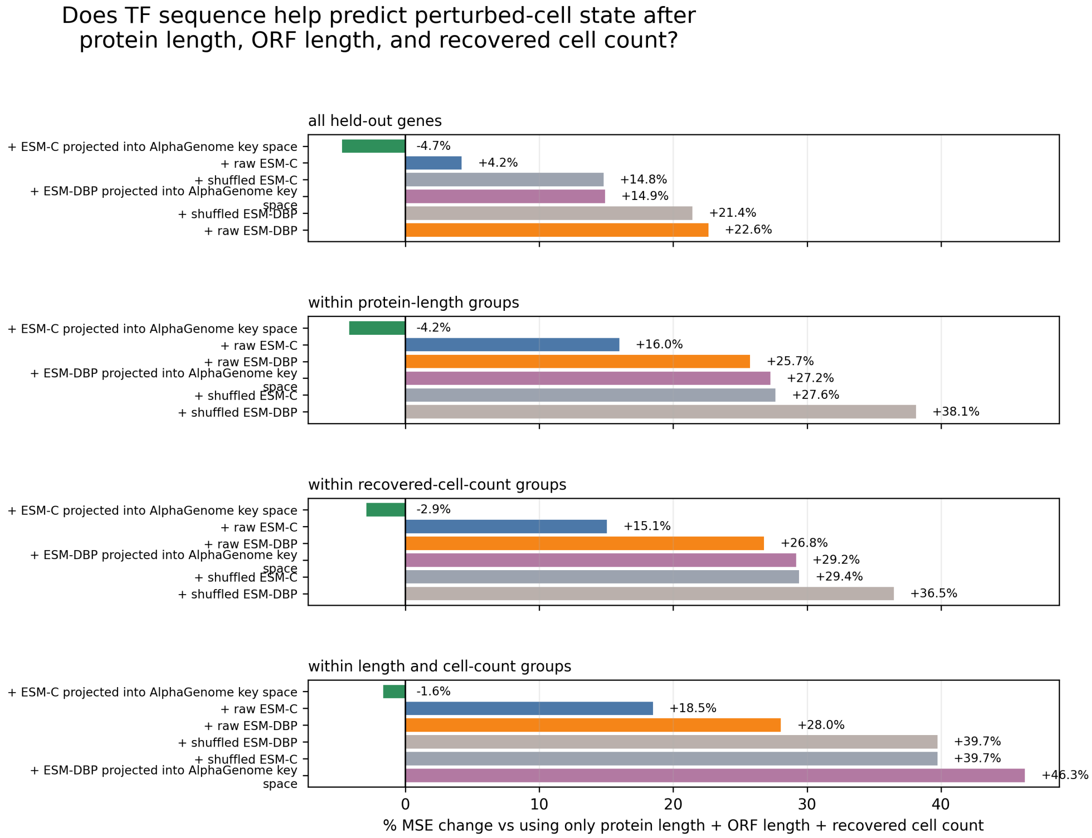

Figure 3: MSE change after adding sequence features to a model that already uses protein length, ORF length, and recovered cell count. ESM-C projected into AlphaGenome key space modestly improves in protein-length groups, recovered-cell-count groups, and joint length-plus-cell-count groups; ESM-DBP is worse in the same comparisons.

6.3 Beta controls retrieval regime

Placeholder. Insert beta sweep results. Report the perturbed-cell state metric, retrieval entropy, effective number of loci, top-k mass, and motif enrichment across beta values.

Figure placeholder: Effect of retrieval temperature β

Plot to create. X-axis: β. Y-axes in separate panels: perturbed-cell state metric, entropy/effective loci, top-k mass, motif enrichment. Expected qualitative regimes: low β diffuse averaging, intermediate β subset retrieval, high β collapse.

Figure 4: Effect of retrieval temperature β. Plot to create. X-axis: β. Y-axes in separate panels: perturbed-cell state metric, entropy/effective loci, top-k mass, motif enrichment. Expected qualitative regimes: low β diffuse averaging, intermediate β subset retrieval, high β collapse.

Interpretation placeholder. Avoid claiming that β is quantitatively equivalent to TF concentration unless dose-calibrated data are available. Safe phrasing: β is an inverse temperature controlling retrieval concentration and may serve as an analog of selectivity or dosage-dependent sharpness.

6.4 Retrieval distributions identify candidate compatible regulatory loci

Placeholder. Insert retrieval diagnostics. For selected TFs, show top-retrieved loci, nearby genes, motif enrichment, accessibility, and expression changes in linked targets.

Table 4: Placeholder TF-specific retrieval case studies.

TF	**Top retrieved loci / annotations**	**Motif or ChIP evidence**	**Perturbation-response interpretation**
`[TF 1]`	[TODO]	[TODO]	[TODO]
`[TF 2]`	[TODO]	[TODO]	[TODO]
`[TF 3]`	[TODO]	[TODO]	[TODO]

Figure placeholder: Interpretable activation patterns over loci

Plot to create. Genome browser-style track or Manhattan-like plot showing retrieval weights over loci for selected TFs and cell-state clusters. Add tracks for accessibility, motif hits, nearby genes, and observed expression changes if available.

Figure 5: Interpretable activation patterns over loci. Plot to create. Genome browser-style track or Manhattan-like plot showing retrieval weights over loci for selected TFs and cell-state clusters. Add tracks for accessibility, motif hits, nearby genes, and observed expression changes if available.

Interpretation placeholder. Use cautious language: high-weight loci are candidate compatible regulatory contexts. They are not validated binding sites unless supported by independent evidence.

6.5 Motif enrichment and external binding validation

Placeholder. Insert motif enrichment analysis. Use matched background loci to control for GC content, accessibility, sequence length, and mappability. Report enrichment for cognate motifs and related TF-family motifs.

Table 5: Placeholder motif enrichment results for top-retrieved loci.

TF / family	**Top-k**	**Cognate motif enrichment**	**Adjusted p-value**	**Matched background**
`[TF 1]`	[TODO]	[TODO]	[TODO]	[TODO]
`[TF 2]`	[TODO]	[TODO]	[TODO]	[TODO]
`[TF 3]`	[TODO]	[TODO]	[TODO]	[TODO]

Interpretation placeholder. If motif enrichment is weak, do not force a mechanistic interpretation. Possible explanations include inadequate memory embeddings, missing cell-state context, non-DNA-binding cofactors, low-quality TF sequence embeddings, or perturbed-cell-state supervision dominated by dataset structure.

6.6 Protein length, ORF length, and recovered-cell-count checks

Predicting TF Atlas perturbed-cell states is strongly affected by protein length, ORF length, and recovered cell count. Longer ORFs and proteins are harder to package and infect in lentiviral libraries, reducing recovered cell counts. Separately, strong responders can reduce recovered cells because cells die or leave the assayed state. These mechanisms interact so that protein length, recovered cell count, and response score are correlated in the observed data. We therefore model protein length, ORF length, and recovered cell count directly rather than treating them as incidental metadata.

Table 6: Checks using protein length, ORF length, and recovered cell count. Lower MSE is better. The matched rows evaluate whether sequence-derived features add signal within groups defined by protein length and recovered cell count.

Control	MSE	Mean cosine similarity	Conclusion
Protein length only	6.6355	0.4114	Strong by itself
Protein length + cell count	6.6020	0.3994	Nearly matches length+cell+ORF
Protein length + ORF length + recovered cell count	6.5996	0.3970	Reference for the sequence-feature comparisons
Cell count only	7.2557	0.3161	Weaker alone than length-based covariates
ESM-C projected into AlphaGenome key space, protein-length groups	6.2714	0.4716	4.2% lower MSE
ESM-C projected into AlphaGenome key space, cell-count groups	6.4074	0.3878	2.9% lower MSE
ESM-C projected into AlphaGenome key space, length+cell groups	6.4603	0.4604	1.6% lower MSE
ESM-DBP projected into AlphaGenome key space, length+cell groups	9.6068	0.3185	Worse

These controls are central to the interpretation. Protein length, recovered cell count, and ORF length explain enough of the perturbed-cell state that raw sequence features alone are not convincing. The strongest result is a small but consistent improvement from adding the ESM-C-derived representation after projection into AlphaGenome key space, including within protein-length groups, recovered-cell-count groups, and joint length-plus-cell-count groups. The safe interpretation is that this sequence representation may add a modest amount of information for predicting the state of the perturbed cell, but the result should be paired with retrieval diagnostics and biological validation before making mechanistic claims.

7 Discussion

7.1 What HopTF adds beyond "slapping on a transformer"

HopTF does not claim novelty from the softmax operation itself. The retrieval layer is mathematically related to attention, and modern Hopfield networks explicitly connect attention to associative memory. The contribution is the biological decomposition: protein sequence provides the query, genomic loci provide the addressable memory, retrieval produces a regulatory-context hypothesis, and a conditional flow model maps that retrieved representation to a predicted cell-state shift. A generic transformer perturbation model may improve expressivity, but it typically does not specify the biological meaning of queries, keys, values, masks, retrieval temperatures, or retrieval diagnostics.

This distinction matters because HopTF exposes a falsifiable intermediate object. The locus activation distribution can be audited against motifs, accessibility, target-gene annotations, ChIP-seq evidence, protein length, ORF length, recovered cell count, and shuffled sequence representations. Predicting the perturbed-cell state alone is not sufficient evidence that the model has learned biologically meaningful regulation. The retrieval distribution is therefore both the mechanism and the liability of the model: it creates interpretability, but it must be validated.

7.2 Associative memory as a modeling lens

The memory in HopTF is the locus-indexed key/value bank, not the transport predictor. The transport predictor is the downstream flow module that uses the retrieved memory state. In associative-memory terms, genomic locus embeddings are stored patterns, the TF isoform embedding is the retrieval cue, and μ_p is the retrieved regulatory-context prototype. This framing connects HopTF to the memory-model side of the course. The model studies how a biological perturbation cue can retrieve information from a large structured memory under weak aggregate supervision.

The model is not a continual-learning algorithm in its current form. However, the explicit memory-bank formulation suggests continual extensions. New cell types, accessibility profiles, perturbation datasets, or genomic annotations could update or augment the regulatory memory while preserving the retrieval-and-transport interface. Robust continual updating would still require mechanisms such as replay, regularization, adapters, or frozen base memories to avoid catastrophic forgetting.

7.3 Biological interpretation and hypothesis generation

HopTF is best interpreted as a hypothesis-generation model. We tolerate false positives more than false negatives if the goal is to nominate candidate compatible loci for follow-up analysis. This changes the evaluation standard. The model does not need every high-weight locus to be causal, but it should enrich for plausible regulatory contexts, avoid obvious confounding by dataset structure, and recover known motifs or target programs when strong prior evidence exists.

A successful result would look like this: predictions of perturbed-cell state are competitive with simple comparisons; retrieval is not collapsed or random; top-retrieved loci are enriched for cognate or family-level motifs; retrieved loci are plausible under accessibility or cluster context; and case studies produce testable regulatory hypotheses. An unsuccessful but still informative result would show that perturbed-cell state prediction is driven by sequence length, TF family, or dataset structure rather than regulatory retrieval. In that case, HopTF would still clarify what additional supervision or cell-state information is needed.

7.4 Limitations

Several limitations should be explicit.

1. **Weak supervision.** The model is trained on aggregate perturbation responses, not locus-level binding or causal regulatory labels. Retrieval weights are therefore not binding probabilities.
2. **Cell-state context.** Fully cell-specific accessibility masking may be unrealistic if accessibility is not observed or accurately inferred. Cluster-specific biases may be more stable but less granular.
3. **Pretrained embedding dependence.** If AlphaGenome or protein embeddings fail to encode the relevant TF-locus compatibility information, the retrieval module may learn shortcuts.
4. **Dataset structure.** Perturbation cell counts, ORF length, TF family identity, baseline expression, and barcode effects may explain perturbed-cell-state metrics unless explicitly controlled.
5. **Computational scale.** Genome-wide retrieval over millions of loci may require approximate retrieval, candidate pruning, hierarchical memory, or region-level prefiltering.
6. **Causal ambiguity.** Even motif-enriched high-retrieval loci may be passengers or correlated contexts rather than causal mediators of the transcriptional response.

7.5 Future directions

Natural extensions include adapter-based task adaptation of AlphaGenome-derived memory slots, hierarchical retrieval from chromosomes to loci, explicit enhancer-gene linking in value vectors, auxiliary motif or ChIP supervision, dose-conditioned retrieval temperature calibration, and continual updates to the memory bank as new cell types and perturbation datasets become available. A stronger biological model would also distinguish direct TF binding, cofactor-mediated regulation, chromatin remodeling, and downstream secondary response programs.

8 Conclusion

HopTF formulates TF perturbation prediction as sequence-conditioned associative retrieval over a genome-indexed regulatory memory. A TF isoform sequence embedding queries AlphaGenome-derived locus embeddings, optionally modulated by cell-state context, and the resulting activation distribution over loci is pooled into a perturbation-conditioning representation μ_p. This representation conditions a flow model that predicts the transport from control to TF-perturbed cell states. The model is end-to-end differentiable through retrieval and response prediction, allowing aggregate perturbation supervision to shape the addressing function even without locus-level labels. The central claim is therefore not that attention alone is novel, but that biologically specified associative retrieval creates an interpretable, falsifiable intermediate between TF sequence and cell-state response.

Placeholder. Replace with final conclusion after experiments. Include one sentence on perturbed-cell state prediction, one sentence on retrieval diagnostics, and one sentence on the strongest remaining limitation.

A Appendix: Suggested Figures

1. **Figure 1: Model overview.** TF sequence query, AlphaGenome locus memory, cell-state bias/mask, retrieval weights, μ_p, conditional flow field.
2. **Figure 2: Flow prediction.** Control, observed perturbed, and predicted perturbed latent distributions for selected TFs.
3. **Figure 3: Beta sweep.** Perturbed-cell-state performance and retrieval concentration across β.
4. **Figure 4: Retrieval tracks.** Top-retrieved loci for selected TFs with motif/accessibility/target-gene annotations.
5. **Figure 5: Length, cell-count, and shuffled-sequence checks.** Performance of length/cell-count inputs and shuffled-sequence comparisons.

B Appendix: Reporting Checklist

- State whether AlphaGenome, ESM-C, and SCARF are frozen, adapter-tuned, or fully fine-tuned.
- State whether values are tied to keys.
- State whether retrieval is unmasked, cluster-masked, or cell-specific.
- Report primary perturbed-cell-state metric and confidence intervals.
- Report retrieval entropy, top-k mass, and beta sensitivity.
- Include at least one comparison that uses protein length, ORF length, and recovered cell count.
- Avoid interpreting retrieval weights as binding sites without external validation.

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